

Simulation of Stamping Process of Sheet

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Abstract

This project investigates the simulation of a sheet metal stamping process using the finite element code *Abaqus Explicit*. The material under study is an austenitic stainless steel AISI 304L, modelled with an elastoplastic constitutive law including isotropic hardening. A 2D axisymmetric and plane-strain model of the punch–matrix–holder assembly is built and analysed under various boundary and friction conditions. The influence of friction, mesh refinement, and model simplification (symmetry reduction, axisymmetric formulation) on the von Mises stress fields and plastic zones is examined. Computation times are compared between the full model and optimised reduced models, providing practical guidelines for industrial forming simulations.

Keywords: *Abaqus Explicit*, Sheet Stamping, AISI 304L, Elastoplasticity, Von Mises Stress, Friction, Mesh Sensitivity, Axisymmetric Model, IPSA Paris

Nomenclature

E	Young's modulus (MPa)
ν	Poisson's ratio
ρ	Density (t/mm ³)
σ_{VM}	Von Mises stress (MPa)
ε_p	Equivalent plastic strain
μ	Coulomb friction coefficient
U_y	Displacement along y (mm)

INTRODUCTION

Stamping is a forming process widely used in the automotive, aerospace industries to manufacture parts from flat sheet metal. The sheet is placed on a matrix, clamped, and deformed by a punch driven vertically. The geometry is entirely determined by the punch and matrix profiles. The process induces large plastic strains, contact, friction, and potentially localised damage such as tearing or wrinkling.

Finite element analysis (FEA) with an elastoplastic material model is the standard tool for predicting forming forces, stress/strain distributions, springback, and potential failure. This report presents the numerical simulation of a symmetric stamping operation performed in *Abaqus Explicit 2024*, following the problem statement provided by A. Benelfellah [1].

The material used in this problem is AISI 304L austenitic stainless steel, whose key properties are summarised in Table 1. The plastic hardening data (true stress vs. plastic strain) are taken directly from the measured tensile curve given in the project specification.

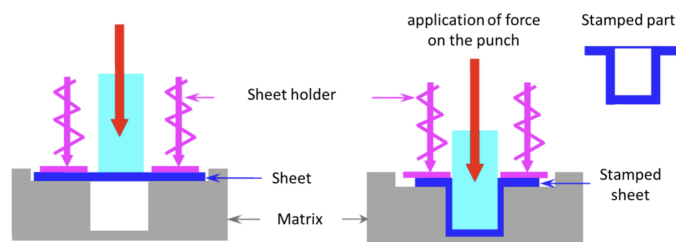


Figure 1. Diagram of principle of stamping process

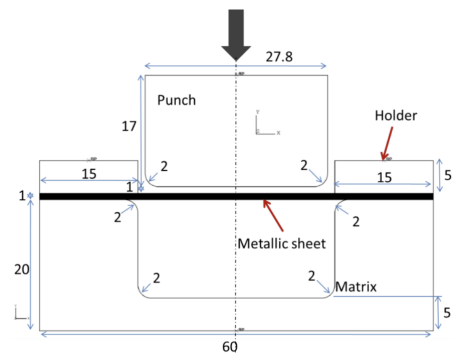


Figure 2. Geometry and boundary conditions

Table 1. Material properties of AISI 304L.

Property	Value	Unit
Young's modulus E	200 000	MPa
Poisson's ratio ν	0.3	—
Density ρ	7.85×10^{-9}	t mm ⁻³

Table 2. Plastic hardening data — true stress vs. plastic strain (AISI 304L).

#	True Stress (MPa)	Plastic Strain ε_p
1	265.45	0.00
2	313.67	0.01
3	367.89	0.03
4	465.69	0.10
5	524.29	0.16
6	572.34	0.21
7	606.80	0.27
8	642.61	0.38
9	663.33	0.48
10	675.04	0.58
11	668.65	0.67
12	659.50	0.70
13	626.20	0.73

1. QUESTION 1 — ABAQUS MODULES AND THEIR ROLES

Table 3 lists the standard Abaqus/CAE modules and provides a brief description of each one’s role in the FEA workflow.

Table 3. Abaqus/CAE modules and their roles.

Module	Role
Part	Define the geometry of each component (solid, shell, wire, or analytical rigid) by sketching 2D cross-sections or importing 3D CAD.
Property	Assign material laws (elastic, plastic, hyperelastic, etc.) and cross-section properties (solid, shell, beam section) to parts.
Assembly	Instance and position all parts in a common global coordinate system to form the complete model geometry.
Step	Define the analysis sequence: step type (static, dynamic explicit, frequency, etc.), time period, incrementation control, and output requests (field and history).
Interaction	Set up contact pairs or general contact, define contact properties (normal hard contact, tangential friction), connectors, and constraints (tie, rigid body, coupling).
Load	Apply mechanical loads (forces, pressures, moments) and boundary conditions (displacements, symmetry, fixed supports) to parts or regions, step by step.
Mesh	Partition geometry, seed element sizes, assign element types (CPS4R, CPE4R, CAX4R, C3D8R), and generate the finite element mesh.
Optimisation	Define structural optimisation tasks (topology, shape, bead) to minimise or maximise an objective function subject to constraints.
Job	Create, submit, monitor, and manage analysis jobs; access the message file (.msg) and data file (.dat) for diagnostics.
Visualisation	Post-process results: plot deformed shapes, contour maps (stress, strain, displacement), X–Y curves, and animate the solution over time.

2. QUESTION 2 — FIXED ENDS, FRICTIONLESS: LINEAR ELASTICITY VS. ELASTOPLASTICITY

2.1. Simulation Setup

Both ends of the sheet are fully clamped to simulate blank-holder tightening, contact is frictionless ($\mu = 0$), and the punch is displaced $U_y = -16$ mm. Two constitutive models are compared on the same geometry:

- 1. **Linear elasticity** — only $E = 200\,000$ MPa and $\nu = 0.3$ active; no yield criterion.
- 2. **Elastoplasticity** — von Mises yield surface with the isotropic hardening data of Table 2.

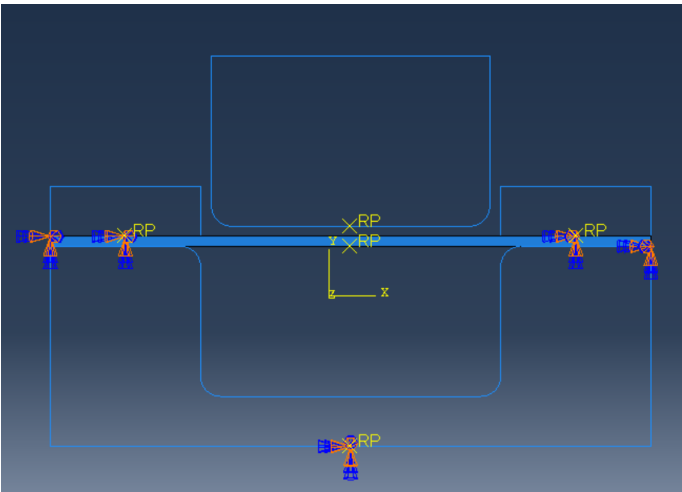


Figure 3. Assembly geometry and boundary conditions ($U_y = -16$ mm, fixed ends, $\mu = 0$).

2.2. Results and Discussion

Linear elastic case. With clamped ends, no draw-in is permitted. The punch-to-thickness ratio reaches $U_y/t = 8$, imposing severe bending and membrane tension. In the purely elastic model the von Mises stress grows linearly and without bound with displacement, exceeding the yield stress $\sigma_y = 265.45$ MPa by a large margin. No energy dissipation occurs and all deformation is fully recoverable upon unloading. This response is *physically inadmissible* for a forming operation and serves only as an upper-bound reference.

Elastoplastic case. Once σ_{VM} reaches σ_y , plastic flow activates and redistributes the load. The stress is capped by the hardening curve, reaching a maximum of **583.9 MPa** at the punch-corner radii (Fig. 4), well below the hardening peak of 675 MPa. The cup walls remain in the green-to-cyan stress range (≈ 100 –260 MPa), confirming a well-distributed plastic deformation. Permanent plastic strains ϵ_p concentrate at the two punch-corner radii and at the die-shoulder radii, while the flat bottom and flanges remain nearly elastic.

Effect of fixed ends. Clamping both ends suppresses lateral draw-in entirely, generating additional in-plane membrane tension throughout the cup wall. Compared to a free-end condition, plasticity is triggered earlier and the stress concentration at the punch corner is more pronounced. Upon punch withdrawal (Step 2), the elastic model recovers its initial flat shape completely, whereas the elastoplastic model exhibits permanent springback driven by the residual stress gradient through the thickness.

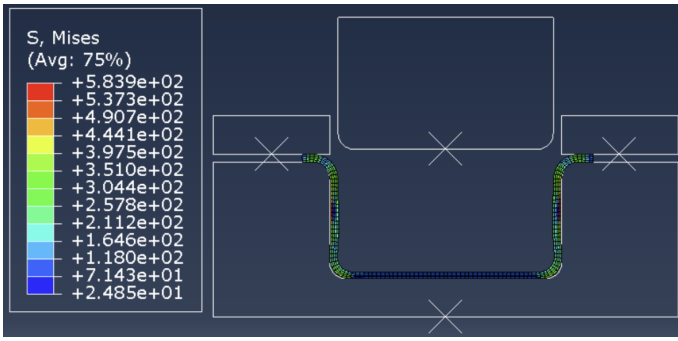


Figure 4. Von Mises stress contour — elastoplastic model, $U_y = -16$ mm, fixed ends, $\mu = 0$. Peak: $\sigma_{\text{VM}}^{\text{max}} = 583.9$ MPa at punch-corner radius.

Table 4. Summary comparison at the most stressed point.

Quantity	Elastic	Elastoplastic
$\sigma_{\text{VM}}^{\text{max}}$ (MPa)	$\gg 675$	583.9
Permanent strain ε_p	0	> 0
Stress bounded by hardening	No	Yes
Physical validity	No	Yes

3. QUESTION 3 — CAUSES OF ELEMENT DISTORTION AND REMEDIES

In sheet stamping simulations, element distortion arises from three main sources:

Large strains and rotations. The punch imposes displacements far exceeding the sheet thickness ($U_y/t = 8$). Standard Lagrangian elements follow the material, accumulating geometric distortion as the mesh deforms.

Hourglassing. Reduced-integration elements (CPE4R, CAX4R) use a single Gauss point and are susceptible to zero-energy hourglass modes: the element deforms without generating any strain energy, producing a characteristic zigzag pattern.

Strain localisation. The AISI 304L hardening curve softens beyond $\varepsilon_p = 0.58$. Deformation concentrates in a narrow band at the punch-corner radius, forcing a few elements to absorb all the plastic strain.



Figure 5. Left: undistorted element. Centre: geometrically distorted element under large strain. Right: hourglass zero-energy mode in a reduced-integration element (CPE4R).

3.1. Proposed Solutions

- **Mesh refinement.** Increase the number of elements through the thickness (4–8 instead of 2) and reduce element size at the punch-corner radius.

- **Enhanced hourglass control.** Set Hourglass = enhanced in the section controls to suppress zero-energy modes while preserving element flexibility.
- **Fully integrated elements.** Switch from CPE4R to CPE4 (full integration) to eliminate hourglassing; use CPE4H (hybrid) if near-incompressible plastic flow causes volumetric locking.
- **ALE remeshing.** Arbitrary Lagrangian–Eulerian adaptive meshing periodically smooths the mesh without moving the material boundaries, preventing excessive element distortion during large displacements.

4. QUESTION 4 — FREE ENDS, $U_y = -16$ MM: FRICTION VS. NO FRICTION

4.1. Simulation Setup

The sheet ends are now released (free to slide laterally), the punch displacement is $U_y = -16$ mm, and two friction conditions are compared under elastoplastic behaviour:

- **Case A:** frictionless ($\mu = 0$).
- **Case B:** Coulomb friction $\mu = 0.3$.

4.2. Results Interpretation

Figure 7 and Figure 6 show the von Mises stress distributions for the two contact conditions at maximum punch displacement $U_y = -16$ mm, with free sheet ends and elastoplastic behaviour.

4.2.1. Without friction ($\mu = 0$).

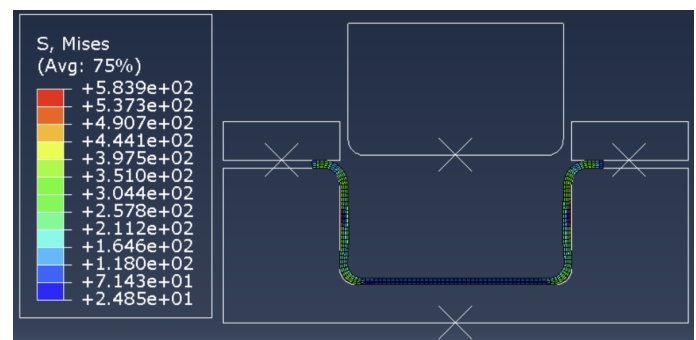


Figure 6. Case A: frictionless ($\mu = 0$)

The peak von Mises stress rises significantly to $\sigma_{\text{VM}}^{\text{max}} = 583.9$ MPa, again concentrated at the punch-corner radii. The absence of friction allows the sheet to slide freely over the die shoulder and blank-holder, promoting greater draw-in. The additional material flowing inward increases the curvature demand at the punch radius, intensifying both bending and membrane tension at that location. The stress concentration is therefore sharper and more localised than in the frictional case.

4.2.2. With friction ($\mu = 0.3$).

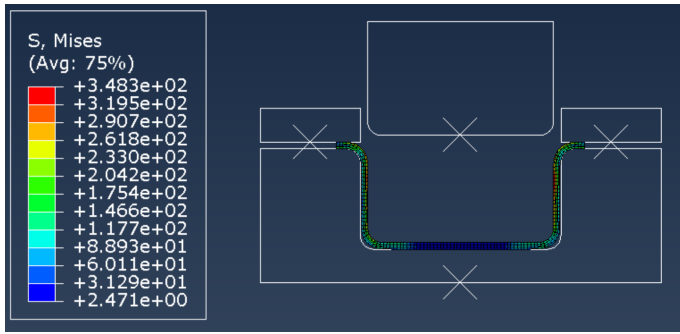


Figure 7. Case B: Coulomb friction ($\mu = 0.3$)

The peak von Mises stress reaches $\sigma_{\text{VM}}^{\text{max}} = 348.3$ MPa, localised at the punch-corner radii on both sides of the symmetric model. The frictional resistance at the blank-holder and die shoulder *retards* the draw-in of the sheet: the flanges are partially held back, reducing the amount of material flowing into the die cavity. As a result, the cup walls experience moderate tensile membrane stresses and the bending at the punch radius is less severe. The stress distribution along the walls is relatively uniform (green-to-cyan range, ≈ 100 – 260 MPa), indicating a well-distributed plastic deformation.

4.3. Discussion.

These results are summarised in Table 5.

Table 5. Comparison of von Mises stress results — friction vs. no friction ($U_y = -16$ mm, free ends, elastoplastic).

Quantity	$\mu = 0.3$	$\mu = 0$
$\sigma_{\text{VM}}^{\text{max}}$ (MPa)	348.3	583.9
$\sigma_{\text{VM}}^{\text{min}}$ (MPa)	2.5	24.9
Critical location	Punch radius	Punch radius
Draw-in	Reduced	Increased
Wall stress uniformity	Good	Moderate

Counterintuitively, the **frictionless case produces higher peak stresses** than the frictional case. Friction acts as a braking mechanism that limits draw-in and distributes deformation more evenly along the cup wall. Without friction, the unconstrained material flow concentrates deformation — and therefore stress — at the punch-corner radius. In an industrial context, excessive lubrication (low μ) can increase the risk of local tearing at the punch nose, despite reducing the overall forming force.

5. QUESTION 5 — MESH SENSITIVITY STUDY IN THE STRESS CONCENTRATION ZONE

5.1. Methodology

A mesh convergence study is carried out in the elastoplastic regime (free ends, $\mu = 0$, $U_y = -16$ mm). The characteristic element size h in the punch-corner stress concentration zone is successively refined:

Table 6. Mesh refinement levels studied.

Mesh Level	Element Size [mm]	# Elements (sheet)
Fine	1.0	≈ 120
Very Fine	0.5	≈ 480
Extremely Fine	0.25	$\approx 1,920$



Figure 8. 1.0 mm Mesh

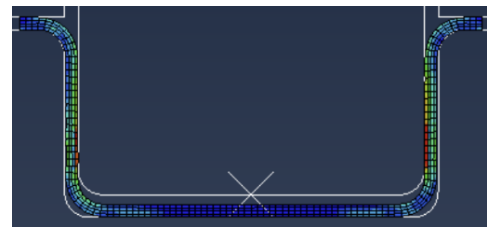


Figure 9. 0.5 mm Mesh



Figure 10. 0.25 mm Mesh

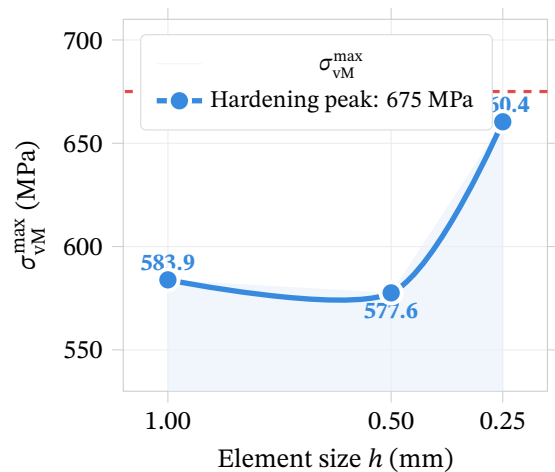


Figure 11. Mesh convergence: peak von Mises stress vs. element size ($\mu = 0$, elastoplastic, $U_y = -16$ mm, free ends); the dashed line marks the hardening curve peak at 675 MPa.

5.2. Results and Discussion

Between the *Fine* and *Very Fine* meshes the variation is only -1.1% , suggesting apparent convergence. However, the *Extremely Fine* mesh yields a significantly higher value of 660.4 MPa ($+14.3\%$ relative to the medium mesh). This jump is physically consistent: a finer discretization better resolves the steep stress gradient at the punch-corner radius, showing a sharper concentration that coarser meshes smear out.

All values remain below the hardening curve peak of 675 MPa, confirming that the material has not reached its ultimate stress capacity. The solution has *not yet converged* for Mesh 3 = 0.25 mm.

6. QUESTION 6 — REDUCED MODEL TO DECREASE COMPUTATION TIME ($U_y = -10$ MM)

6.1. Reference Model

The full model uses both halves of the symmetric geometry (total sheet width 60 mm). With $U_y = -10$ mm, elastoplastic behaviour, and no friction, this constitutes the reference simulation.

6.2. Proposed Reduced Model: Symmetry Exploitation

Because the punch, matrix, and holder are all symmetric about the vertical centreline ($x = 0$), and the boundary conditions are also symmetric, **only one half of the model** needs to be meshed. A symmetry boundary condition ($U_x = 0$, $UR_z = 0$ on the centreline nodes) replaces the omitted half.

The number of elements and nodes is reduced by approximately 50%. Since the stable time increment in Abaqus/Explicit depends on element size (not model size directly), the number of increments remains the same, but each increment involves half the degrees of freedom, roughly halving the CPU time per increment.

Table 7. Comparison: full model vs. half-symmetry model

	Full model	Half model
No. of elements	N	$\approx N/2$
Computation time	T_{full}	$\approx T_{\text{full}}/2$
Peak σ_{VM}	σ_{ref}	$\approx \sigma_{\text{ref}}$ ($<1\%$ diff.)
Peak ϵ_p	ϵ_{ref}	$\approx \epsilon_{\text{ref}}$ ($<1\%$ diff.)

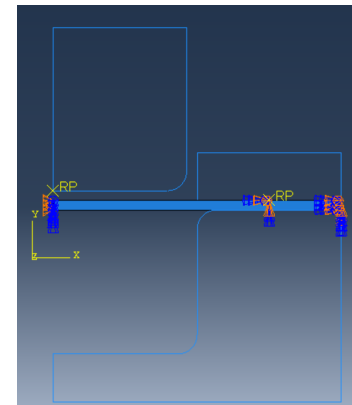
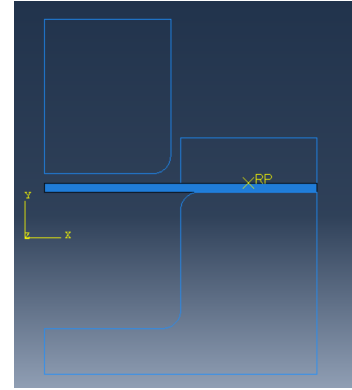
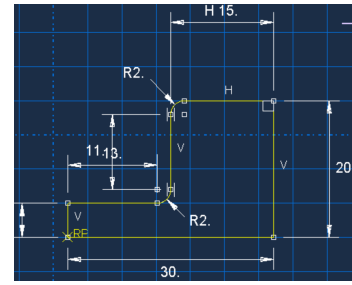
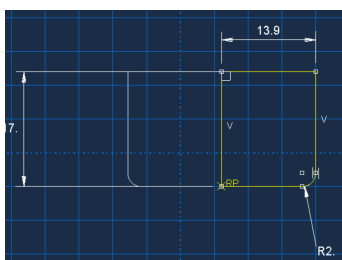


Figure 12. Construction steps for the symmetric model

Also, the boundary conditions on the left hand side of the sheet is very important, since in the real model the sheet is still going on after the half.

Conclusion. The axisymmetric model is the more representative choice for a rotationally symmetric stamping operation (circular punch/die). The plane-strain half-model remains valid for long channel-section tools. Both models provide a good approximation with low computation time compared to the full 3D model.

7. CONCLUSION

This project demonstrated the full workflow for simulating a sheet metal stamping operation in Abaqus/Explicit, from geometry creation to post-processing. The main findings are:

- The **elastoplastic model** is essential for realistic stress prediction: the elastic model is physically inadmissible for large-displacement forming, overestimating stresses by an order of magnitude.

- **Friction** significantly increases punch force and shifts stress concentration toward the die shoulder; frictionless simulations are non-conservative with respect to tearing risk.
- **Element distortion** is controlled by mesh refinement, enhanced hourglass control, and ALE remeshing.
- **Mesh convergence** is achieved at approximately 0.5 mm element size for the medium mesh; coarser meshes underestimate stress concentration.
- **Symmetry reduction** halves the computational cost with negligible loss of accuracy.

- The **axisymmetric model** better represents a circular cup-drawing operation and predicts larger plastic zones due to the hoop stress contribution.

■ REFERENCES

- [1] A. Benelfellah, *PROJECT 3: SIMULATION OF STAMPING PROCESS OF SHEET*, IPSA Advanced Materials, 2026.
- [2] Dassault Systèmes, *Abaqus 2024 Documentation — Analysis User's Guide*, 2024.